

LUMI Prototype: Connection Table for Developer

Board: GOOUUU ESP32-S3-CAM V1.5 (ESP32-S3 N16R8). The OV3660 camera is built into the board and needs no wiring (it uses GPIO 4 to 13 and 15 to 18).

1. Shared rails

The board has one GND, one 3V3 and one 5V header pin. Each is joined to a rail, and every part connects to the rails.

Rail	Source
GND rail	ESP32 GND pin
3V3 rail	ESP32 3V3 pin
5V rail	ESP32 5V pin (fed from the 5 V boost converter when on battery)

2. Connections: from, to

From	To	Notes
OLED GND	GND rail	Pin order on the module: GND, VCC, SCL, SDA
OLED VCC	3V3 rail	3.3 V only
OLED SCL	GPIO 47	I2C clock
OLED SDA	GPIO 21	I2C data
Mic (INMP441) VDD	3V3 rail	Never 5 V
Mic GND	GND rail	
Mic L/R	GND rail	Selects the left channel
Mic SCK	GPIO 41	I2S bit clock, shared with amp BCLK
Mic WS	GPIO 42	I2S word select, shared with amp LRC
Mic SD	GPIO 1	I2S data into the ESP32
Amp (MAX98357A) Vin	5V rail	
Amp GND	GND rail	
Amp BCLK	GPIO 41	Shared with mic SCK
Amp LRC	GPIO 42	Shared with mic WS
Amp DIN	GPIO 14	I2S data out of the ESP32
Amp GAIN	not connected	
Amp SD	not connected	
Speaker (2 W), two wires	Amp's two screw terminals	+ and - either way round; no GPIO
L298N 12V terminal	5V rail	Motor supply from the 5 V rail
L298N 5V terminal	5V rail	Logic supply
L298N GND terminal	GND rail	Common ground
L298N IN1	GPIO 38	Left motor, PWM
L298N IN2	GPIO 39	Left motor, PWM
L298N IN3	GPIO 40	Right motor, PWM
L298N IN4	GPIO 48	Right motor, PWM; also drives the on-board RGB LED
L298N ENA and ENB jumper caps	Keep ON	Motor speed is PWM on IN1 to IN4, so the enables must stay on
L298N 5V-enable jumper cap (next to the screw terminals)	Remove	Logic 5 V is supplied at the 5V terminal
Left motor	L298N OUT1 and OUT2	Swap in firmware if it spins backward

From	To	Notes
Right motor	L298N OUT3 and OUT4	Swap in firmware if it spins backward

3. GPIO summary

GPIO	Function
1	Mic SD (I2S data in)
14	Amp DIN (I2S data out)
21	OLED SDA
38	L298N IN1
39	L298N IN2
40	L298N IN3
41	I2S BCLK (mic SCK and amp BCLK)
42	I2S WS (mic WS and amp LRC)
47	OLED SCL
48	L298N IN4
4-13, 15-18	OV3660 camera (fixed on the board)

Reserved, do not use: 35, 36, 37 (octal PSRAM), 19 and 20 (USB), 43 and 44 (serial/CH340), 0, 3, 45, 46 (boot/strapping pins).

4. Power (summary)

Two 18650 Li-ion cells in parallel go to the charger module (B+ and B-). The charger OUT+ goes through a switch to the boost converter IN+, and OUT- to IN-. The boost output is set to 5.0 V and feeds the ESP32 5V and GND pins. The battery never connects to the ESP32 3V3 pin. Use USB or battery, never both, and charge with the switch OFF.

5. Open items

- The OLED controller (SSD1306 or SH1106) is unconfirmed.
- The speaker impedance and the battery protection are unconfirmed.
- The mic, amp, speaker and battery chain have not been tested yet.